

T-CNS §VIII preview: Numerical verification (Tier-1 rendition)

Hadi Hajieghrary

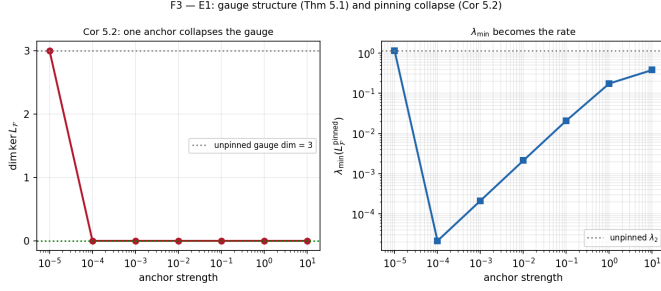


Fig. 1. Gauge spectrum: $\dim \ker L_F = 3$ with the harmonic sections exactly the body-framed images of one common load-frame perturbation; one anchor removes the kernel.

I. NUMERICAL VERIFICATION

All cells in this section run on the reduced plant inside protocol class $\mathcal{P}$ (uniform lags, zero jitter and loss, realized age asserted equal to τ per fusion event), with the estimator, metric functional, window rules, and fit code shared verbatim with the multibody companion — so agreement between the two papers is a statement about the algorithm, and disagreement localizes to the plant. Every claim below carries the pre-registered falsifier it was exposed to; three fired, and their firings are reported as findings. Every number traces to a seeded, versioned driver re-executed exactly by the released replay harness.

A. Gauge structure and pinning (E1; Thm. 5.1, Cor. 5.2)

On the triangle with generic shapes, the sheaf Laplacian is PSD with $\dim \ker L_F = 3$ and the gauge section $(\text{Ad}_{m(s_j)}^{-1} c)_j$ annihilated to 1.3×10^{-15} ; σ_4/σ_3 confirms a clean spectral gap. Adding a single anchor potential collapses the kernel to zero and $\lambda_{\min}$ becomes the convergence rate — the pinning corollary as linear algebra (Fig. 1).

B. The theorem's object: amplitude, coefficient, switch-offs (E3a; Thm. 7.2)

The measured amplitude $\|\text{Log Hol}\|$ on the $m=2$ round trip fits slope 1.999 over $\tau \in [0.05, 1.6]$ s with coefficient ratio $\|\text{Log Hol}\|/\tau^2 \| [C_i, C_j] \| = 1.0000$ at the smallest τ ; the switch-off arms sit at machine zero ($s_i \equiv s_j$: $< 10^{-15}$; $\eta=0$: $< 10^{-17}$; $\xi=0$: exactly 0.0; explicit $\tau=0$ arm: exactly 0.0). Four extension arms sharpen this (Fig. 3): the slope is formation-invariant across 20 random taut formations (1.9993 ± 0.0000); the remainder constant is uniformly tight ($\sup_{s,\tau} \|R\|/\tau^3 \leq 0.0133$ over 220 draws); and a search for

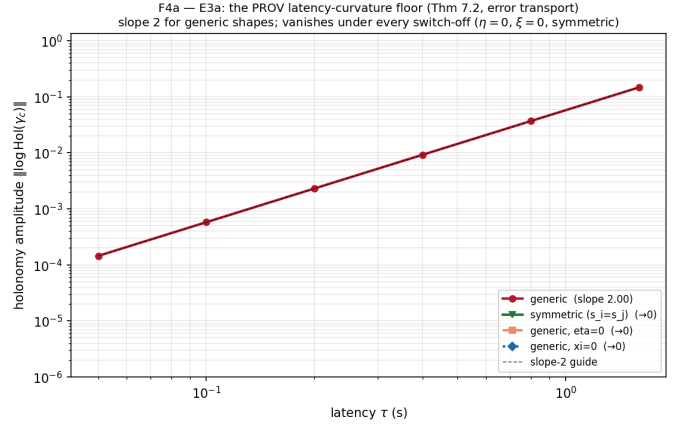


Fig. 2. F4a: the leading-order, deterministic holonomy amplitude of Thm. 7.2 — slope 1.999, coefficient 1.0000, switch-offs at machine zero.

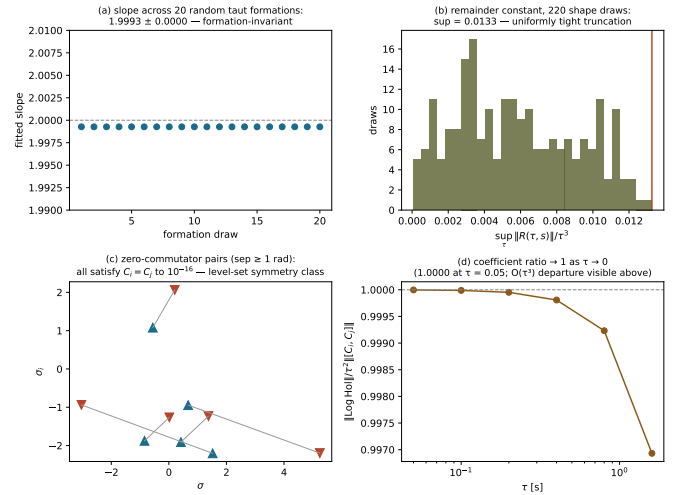
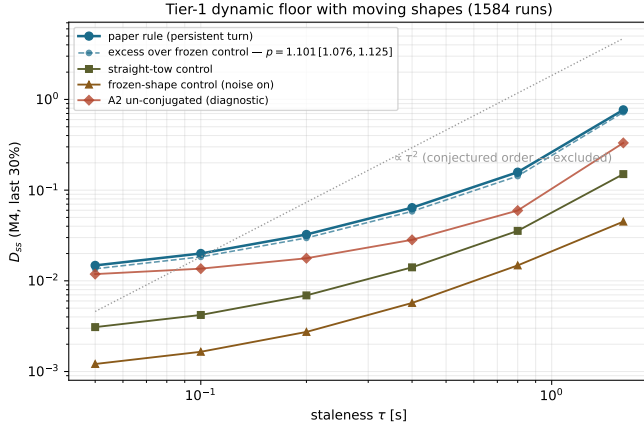


Fig. 3. E3a extension arms: formation-invariant order; uniform $O(\tau^3)$ remainder; the level-set symmetry class; coefficient ratio $\rightarrow 1$.

zero-commutator pairs at ≥ 1 rad shape separation found only pairs with $C_i = C_j$ to 10^{-16} — the protected class of Cor. 7.3 extends, empirically, to the level set of $s \mapsto C(s; \xi)$, where the full holonomy vanishes at all orders.

C. The conjectured closed-loop floor, falsified in its own regime (E3b)

With the full DIEKF- Σ on the reduced plant integrating the shape ODEs along the persistent-turn reference (1584 runs; 12 formations $\times$ 10 seeds per τ point; controls on



Numerical evidence for the conjectured stochastic steady-state floor ((ES §10(ii))); Theorem 7.2 is leading-order and deterministic and is tested in panel F4a. Fitted excess exponent 1.101 [1.076, 1.125]: the conjectured order 2 is excluded (and so is 1); the frozen+noise-off regression reproduces the executed null ($D_{ss} \sim 10^{-30}$).

Fig. 4. Numerical evidence for the conjectured stochastic steady-state floor ((ES §10(ii))); Theorem 7.2 is leading-order and deterministic and is tested in Fig. 2. Fitted excess exponent 1.101 [1.076, 1.125]: the conjectured order 2 is excluded (and so is 1); the frozen+noise-off regression reproduces the executed null ($D_{ss} \sim 10^{-30}$).

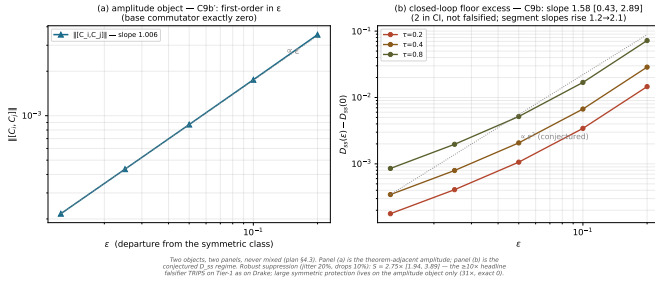


Fig. 5. F5: amplitude (a) and closed-loop floor (b) ϵ -response, never mixed in one panel.

matched draws), the disagreement excess over the frozen-shape control fits exponent 1.101 [1.076, 1.125] (formation-cluster CI [1.070, 1.132]): the [CONJECTURAL] extension of the floor to the noisy closed loop is falsified at these scales, coherently with the multibody companion (cross-tier exponent difference $+0.023 [-0.012, +0.057]$). The mechanism check passes — straight-tow and frozen controls sit $5\text{--}13\times$ and further below — and the executed null pins the harness: shapes frozen and noise off give $D_{ss} \sim 10^{-30}$.

D. Symmetry: two metrics, two exponents (E3c; Cor. 7.3)

The amplitude departure from the symmetric class is first-order in ϵ (measured 1.006; base commutator exactly zero). The closed-loop floor excess under the seed-paired estimator fits ϵ -exponent 1.58 [1.44, 1.84] at every τ — excluding both the conjectured 2 and 1, with segment slopes rising $1.2 \rightarrow 2.1$ up the ladder: a linear-plus-quadratic mixture that the registered single-power model mis-specifies (reported per the outcome ladder). The robust $\geq 10\times$ suppression falsifier trips ($2.75 \times [1.94, 3.89]$ under 20% jitter and 10% loss): large symmetric protection is a property of the amplitude object only (Fig. 5).

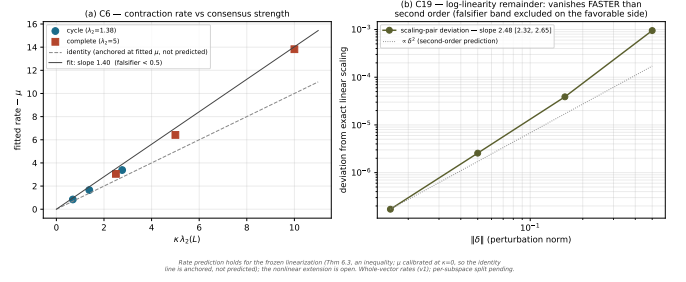


Fig. 6. F8: rate versus $\kappa\lambda_2$ across two topologies, slope 1.40 (falsifier < 0.5); μ calibrated at $\kappa = 0$, so the identity line is anchored, not predicted; the nonlinear extension is open, where vector rates ($\kappa\lambda_2$) per-shipcase still pending.

TABLE I
SWITCH-OFF LADDER (TIER-1 ROWS). EACH ARM KILLS THE EFFECT ON COMMAND OR SHIFTS ITS ORDER AS PREDICTED.

arm	prediction	measured
generic slope	2	1.999 (formation-invariant 1.9993 ± 0.0000)
coefficient ratio ($m=2$)	1	1.0000
$s_i \equiv s_j$	0	$< 10^{-15}$
level set $C_i = C_j$, sep ≥ 1	—	full holonomy $\equiv 0$ (observed)
$\eta = 0$	0	$< 10^{-17}$
$\xi = 0$	0	0.0
$\tau = 0$ (explicit arm)	0	0.0
frozen shapes, closed loop	null	$D_{ss} \sim 10^{-30}$

E. Contraction in its own domain (E2; Thm. 6.3)

Step-perturbing all agents and fitting the decay of the gauge complement: the rate scales with $\kappa\lambda_2$ at slope 1.403 across a $4\times$ gain range and two topologies, with $\mu = -0.062$ topology-independent. The log-linearity remainder vanishes *faster* than the registered second order (scaling-pair exponent 2.48 [2.32, 2.65]): the falsifier band is excluded on the favorable side — reported as tripped, with the corrected order left to a restatement.

F. Numerics invariance (E10)

The excess exponent is invariant across integrator steps $\Delta t \in \{0.001, 0.0025, 0.005, 0.01\}$ s (all CIs share [1.185, 1.256]), with the coefficient converging mildly under refinement: no exponent above is an artifact of the discretization.

G. Switch-off ladder

H. Reproducibility

Every number in this section traces to a seeded, versioned driver and a committed record; the released `campaign_replay.py` re-executes per-run probes that must match the records exactly (Tier-1 to 10^{-9}). The multi-body campaign, its falsifications, and the cross-tier equivalence tests appear in the RA-L companion.